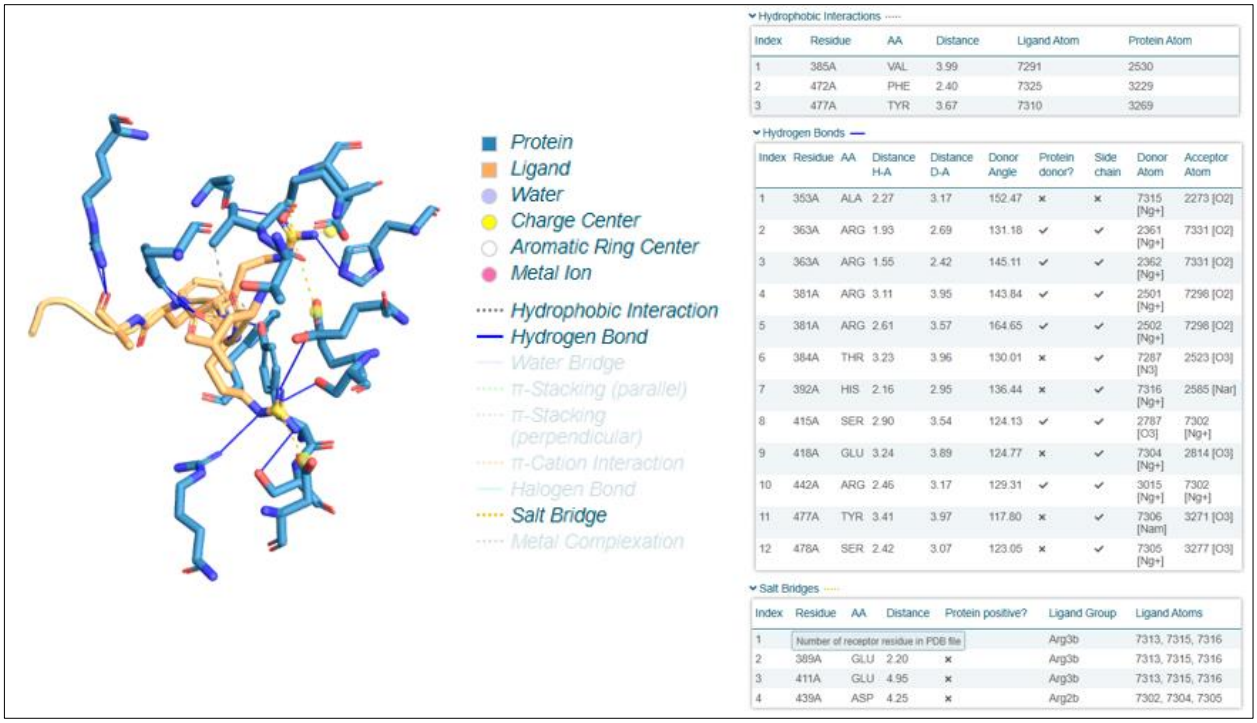
APÉNDICE O

RESULTADOS OBTENIDOS DE PLIP

Secuencia 7: LRRFALA



Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	385A	VAL	3.99	7291	2530
2	472A	PHE	2.40	7325	3229
3	477A	TYR	3.67	7310	3269

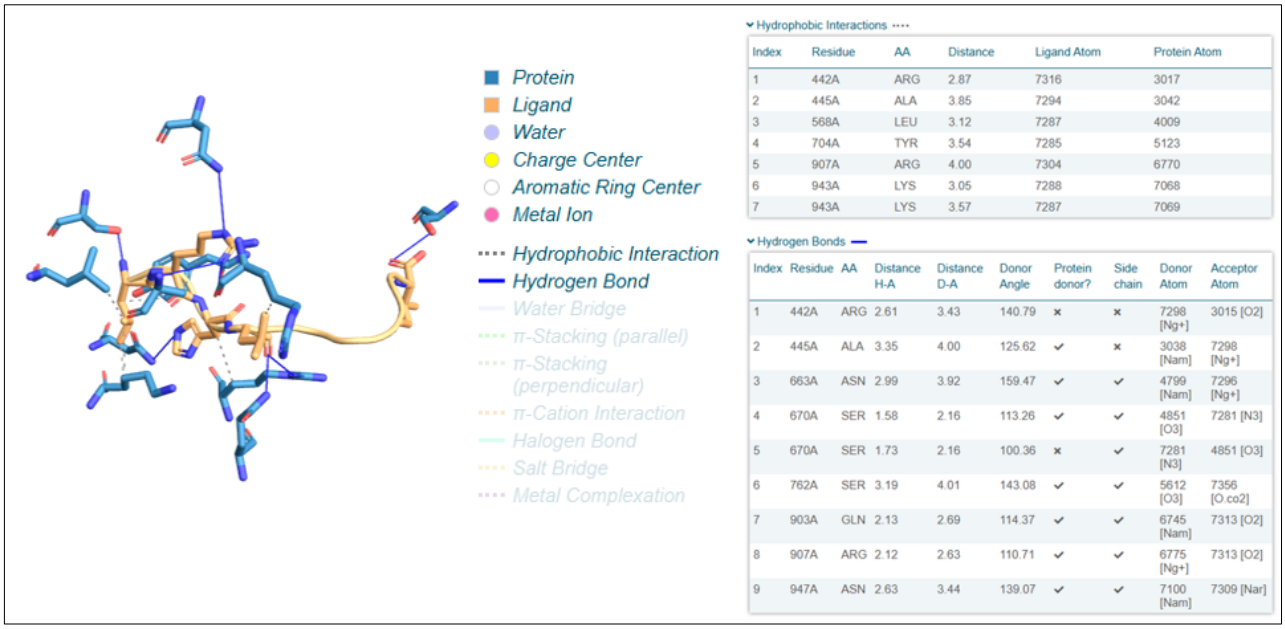
Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	353A	ALA	2.27	3.17	152.47	×	×	7315 [Ng+]	2273 [O2]
2	363A	ARG	1.93	2.69	131.18	✓	✓	2361 [Ng+]	7331 [O2]
3	363A	ARG	1.55	2.42	145.11	✓	✓	2362 [Ng+]	7331 [O2]
4	381A	ARG	3.11	3.95	143.84	✓	✓	2501 [Ng+]	7298 [O2]
5	381A	ARG	2.61	3.57	164.65	✓	✓	2502 [Ng+]	7298 [O2]
6	384A	THR	3.23	3.96	130.01	×	✓	7287 [N3]	2523 [O3]
7	392A	HIS	2.16	2.95	136.44	×	✓	7316 [Ng+]	2585 [Nar]
8	415A	SER	2.90	3.54	124.13	✓	✓	2787 [O3]	7302 [Ng+]
9	418A	GLU	3.24	3.89	124.77	×	✓	7304 [Ng+]	2814 [O3]
10	442A	ARG	2.46	3.17	129.31	✓	✓	3015 [Ng+]	7302 [Ng+]
11	477A	TYR	3.41	3.97	117.80	×	✓	7306 [Nam]	3271 [O3]
12	478A	SER	2.42	3.07	123.05	×	✓	7305 [Ng+]	3277 [O3]

Salt Bridges

Index	Residue	AA	Distance	Protein positive?	Ligand Group	Ligand Atoms
1	Number of receptor residue in PDB file				Arg3b	7313, 7315, 7316
2	389A	GLU	2.20	×	Arg3b	7313, 7315, 7316
3	411A	GLU	4.95	×	Arg3b	7313, 7315, 7316
4	439A	ASP	4.25	×	Arg2b	7302, 7304, 7305

Secuencia 12: LRHLMLRMI



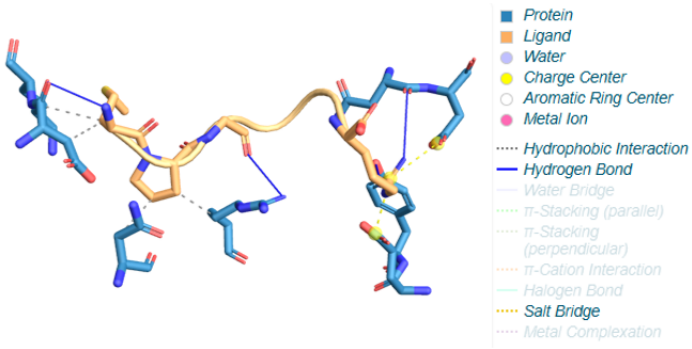
Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	442A	ARG	2.87	7316	3017
2	445A	ALA	3.85	7294	3042
3	568A	LEU	3.12	7287	4009
4	704A	TYR	3.54	7285	5123
5	907A	ARG	4.00	7304	6770
6	943A	LYS	3.05	7288	7068
7	943A	LYS	3.57	7287	7069

Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	442A	ARG	2.61	3.43	140.79	×	×	7298 [Ng+]	3015 [O2]
2	445A	ALA	3.35	4.00	125.62	✓	×	3038 [Nam]	7298 [Ng+]
3	663A	ASN	2.99	3.92	159.47	✓	✓	4799 [Nam]	7296 [Ng+]
4	670A	SER	1.58	2.16	113.26	✓	✓	4851 [O3]	7281 [N3]
5	670A	SER	1.73	2.16	100.36	×	✓	7281 [N3]	4851 [O3]
6	762A	SER	3.19	4.01	143.08	✓	✓	5612 [O3]	7356 [O.co2]
7	903A	GLN	2.13	2.69	114.37	✓	✓	6745 [Nam]	7313 [O2]
8	907A	ARG	2.12	2.63	110.71	✓	✓	6775 [Ng+]	7313 [O2]
9	947A	ASN	2.63	3.44	139.07	✓	✓	7100 [Nam]	7309 [Nar]

Secuencia 19: MPGMLR



Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	365A	ASN	3.90	7303	2382
2	381A	ARG	3.04	7302	2503
3	826A	ASP	3.74	7294	6133
4	827A	LYS	3.76	7294	6142

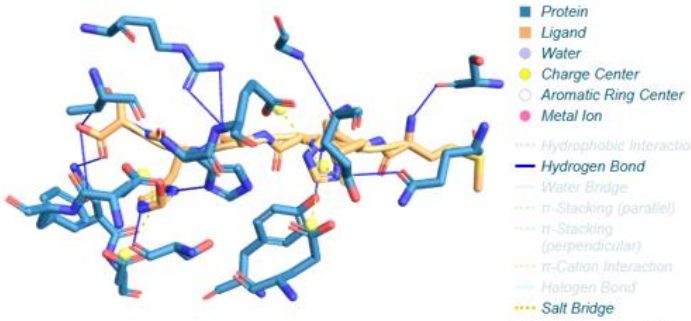
Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	381A	ARG	3.04	3.59	116.61	✓	✓	2507 [Ng+]	7308 [O2]
2	419A	TYR	2.02	2.90	148.28	×	✓	7335 [Ng+]	2833 [O3]
3	438A	ASN	3.24	3.92	127.74	×	×	7334 [Ng+]	2980 [O2]
4	826A	ASP	2.48	3.41	150.54	×	×	7290 [N3]	6132 [O2]

Salt Bridges

Index	Residue	AA	Distance	Protein positive?	Ligand Group	Ligand Atoms
1	418A	GLU	3.46	×	Arg6b	7332, 7334, 7335
2	439A	ASP	4.02	×	Arg6b	7332, 7334, 7335

Secuencia 20: MHLAHA



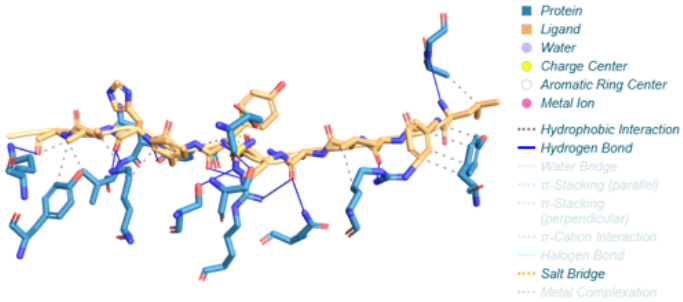
Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	213A	GLN	3.08	3.49	106.85	×	✓	7305 [Nar]	1196 [O2]
2	352A	GLY	3.21	3.89	127.13	✓	×	2272 [Nam]	7301 [O2]
3	381A	ARG	2.03	2.98	164.05	✓	✓	2507 [Ng+]	7319 [O2]
4	381A	ARG	2.62	3.42	137.88	✓	✓	2508 [Ng+]	7319 [O2]
5	384A	THR	3.01	3.93	158.95	✓	✓	2529 [O3]	7336 [O.co2]
6	388A	HIS	3.05	3.62	118.22	×	✓	7328 [Npl]	2557 [N2]
7	415A	SER	1.90	2.56	121.90	×	×	7330 [Npl]	2791 [O2]
8	419A	TYR	3.31	4.01	132.80	✓	✓	2833 [O3]	7336 [O.co2]
9	438A	ASN	2.16	2.92	132.75	✓	✓	2983 [Nam]	7334 [O.co2]
10	477A	TYR	2.53	3.22	127.10	×	✓	7308 [Nam]	3277 [O3]
11	895A	SER	2.80	3.73	151.56	×	×	7290 [N3]	6686 [O2]

Salt Bridges

Index	Residue	AA	Distance	Protein positive?	Ligand Group	Ligand Atoms
1	355A	GLU	3.74	×	His2b	7305, 7307
2	389A	GLU	4.72	×	His2b	7305, 7307
3	411A	GLU	3.27	×	His2b	7305, 7307
4	418A	GLU	2.23	×	His5b	7328, 7330
5	439A	ASP	4.59	×	His5b	7328, 7330

Secuencia 21: LFAVQRYFHIAL



Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	351A	ALA	3.73	7295	2271
2	442A	ARG	3.42	7319	3017
3	477A	TYR	3.18	7306	3270
4	477A	TYR	3.08	7304	3271
5	477A	TYR	3.51	7303	3273
6	477A	TYR	3.32	7302	3275
7	706A	LYS	3.91	7379	5153
8	706A	LYS	3.16	7360	5153
9	715A	TYR	3.66	7380	5233
10	719A	LYS	3.37	7380	5261
11	868A	ASN	3.90	7345	6470
12	869A	ASN	3.85	7368	6478
13	870A	VAL	3.57	7386	6487

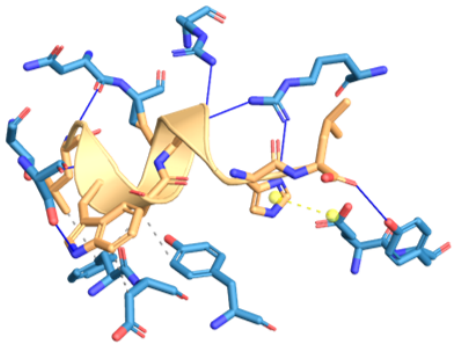
Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	352A	GLY	3.03	3.88	144.49	✔	×	2272 [Nam]	7290 [N3]
2	442A	ARG	2.10	2.50	101.79	✔	✔	3022 [Ng+]	7312 [O2]
3	700A	SER	2.28	3.23	163.50	×	✔	7339 [Ng+]	5103 [O3]
4	703A	SER	2.46	3.24	137.81	✔	✔	5123 [O3]	7337 [Ng+]
5	706A	LYS	1.29	2.22	148.18	✔	✔	5155 [N3+]	7367 [O2]
6	719A	LYS	3.17	4.03	143.28	✔	✔	5263 [N3+]	7385 [O2]
7	868A	ASN	2.60	3.12	113.22	×	×	7364 [Nam]	6469 [O2]
8	870A	VAL	3.06	3.95	150.14	✔	×	6482 [Nam]	7367 [O2]
9	903A	GLN	3.21	3.75	116.27	✔	✔	6754 [Nam]	7324 [O2]
10	907A	ARG	2.93	3.35	106.86	✔	✔	6785 [Ng+]	7344 [O2]
11	907A	ARG	2.82	3.61	137.39	✔	✔	6784 [Ng+]	7324 [O2]

Salt Bridges

Index	Residue	AA	Distance	Protein positive?	Ligand Group	Ligand Atoms
1	696A	GLU	4.01	×	Arg6b	7337, 7339, 7340

Secuencia 23: LLWALMQHL



Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	351A	ALA	0.83	42	2351
2	472A	PHE	3.22	35	3312
3	472A	PHE	3.90	8	3313
4	473A	ASP	3.98	28	3323
5	477A	TYR	3.53	35	3354

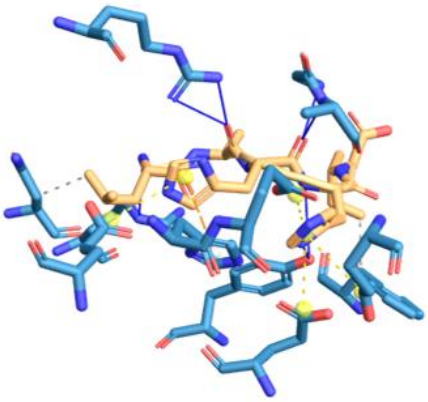
Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	350A	ASN	2.72	3.43	126.89	×	×	1 [N3]	2342 [O2]
2	363A	ARG	2.77	3.57	138.24	✔	✔	2448 [Ng+]	39 [O2]
3	381A	ARG	2.88	3.60	130.76	✔	✔	2588 [Ng+]	39 [O2]
4	381A	ARG	2.63	3.58	162.53	✔	✔	2587 [Ng+]	68 [N2]
5	419A	TYR	3.50	3.93	110.78	✔	✔	2913 [O3]	74 [O.co2]
6	897A	SER	3.44	3.81	104.98	×	✔	17 [Nam]	6785 [O3]
7	899A	SER	2.03	2.98	164.58	✔	✔	6802 [O3]	27 [N2]

Salt Bridges

Index	Residue	AA	Distance	Protein positive?	Ligand Group	Ligand Atoms
1	418A	GLU	3.85	×	His8b	68, 70

Secuencia 60: LAHHI



Protein
Ligand
Water
Charge Center
Aromatic Ring Center
Metal Ion
Hydrophobic Interaction
Hydrogen Bond
Water Bridge
π-Stacking (parallel)
π-Stacking (perpendicular)
π-Cation Interaction
Halogen Bond
Salt Bridge
Metal Complexation

Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	438A	ASN	3.04	7288	2981
2	472A	PHE	2.92	7321	3232

Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	352A	GLY	1.83	2.54	126.37	✓	×	2272 [Nam]	7297 [O2]
2	353A	ALA	2.56	3.37	139.57	✓	×	2276 [Nam]	7297 [O2]
3	381A	ARG	2.44	3.18	131.27	✓	✓	2507 [Ng+]	7292 [O2]
4	381A	ARG	1.80	2.71	153.71	✓	✓	2508 [Ng+]	7292 [O2]
5	442A	ARG	3.03	3.87	141.19	×	✓	7281 [N3]	3019 [Ng+]
6	477A	TYR	2.90	3.57	126.06	×	✓	7304 [Nam]	3277 [O3]
7	477A	TYR	2.37	3.10	129.86	×	✓	7311 [Nar]	3277 [O3]

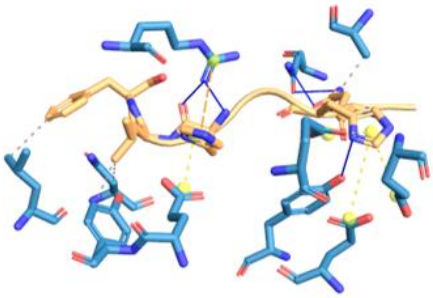
π-Cation Interactions

Index	Residue	AA	Distance	Offset	Protein charged?	Ligand Group	Ligand Atoms
1	388A	HIS	4.01	1.98	×	HIS3B	7301, 7303

Salt Bridges

Index	Residue	AA	Distance	Protein positive?	Ligand Group	Ligand Atoms
1	355A	GLU	4.16	×	His4b	7311, 7313
2	389A	GLU	3.67	×	His4b	7311, 7313
3	411A	GLU	3.80	×	His4b	7311, 7313
4	418A	GLU	3.95	×	His3b	7301, 7303

Secuencia 66: LQHFHLF



Protein
Ligand
Water
Charge Center
Aromatic Ring Center
Metal Ion
Hydrophobic Interaction
Hydrogen Bond
Water Bridge
π-Stacking (parallel)
π-Stacking (perpendicular)
π-Cation Interaction
Halogen Bond
Salt Bridge
Metal Complexation

Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	351A	ALA	3.63	7311	2271
2	419A	TYR	3.42	7345	2830
3	434A	LEU	3.89	7356	2953
4	438A	ASN	3.86	7345	2981

Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	381A	ARG	2.29	3.09	137.18	✓	✓	2507 [Ng+]	7331 [O2]
2	381A	ARG	2.02	2.96	158.55	×	✓	7328 [Nam]	2507 [Ng+]
3	477A	TYR	3.11	3.97	154.35	✓	✓	3277 [O3]	7314 [N2]
4	900A	ASN	2.81	3.70	150.65	×	✓	7305 [Nam]	6730 [O2]
5	900A	ASN	2.98	3.66	127.16	✓	✓	6729 [Nam]	7306 [O2]

π-Cation Interactions

Index	Residue	AA	Distance	Offset	Protein charged?	Ligand Group	Ligand Atoms
1	381A	ARG	4.62	0.76	✓	Aromatic	7333, 7334, 7335, 7336, 7337

Salt Bridges

Index	Residue	AA	Distance	Protein positive?	Ligand Group	Ligand Atoms
1	355A	GLU	4.77	×	His3b	7314, 7316
2	389A	GLU	5.02	×	His3b	7314, 7316
3	411A	GLU	5.11	×	His3b	7314, 7316
4	418A	GLU	3.19	×	His5b	7335, 7337

Secuencia 73: MAALRHLL



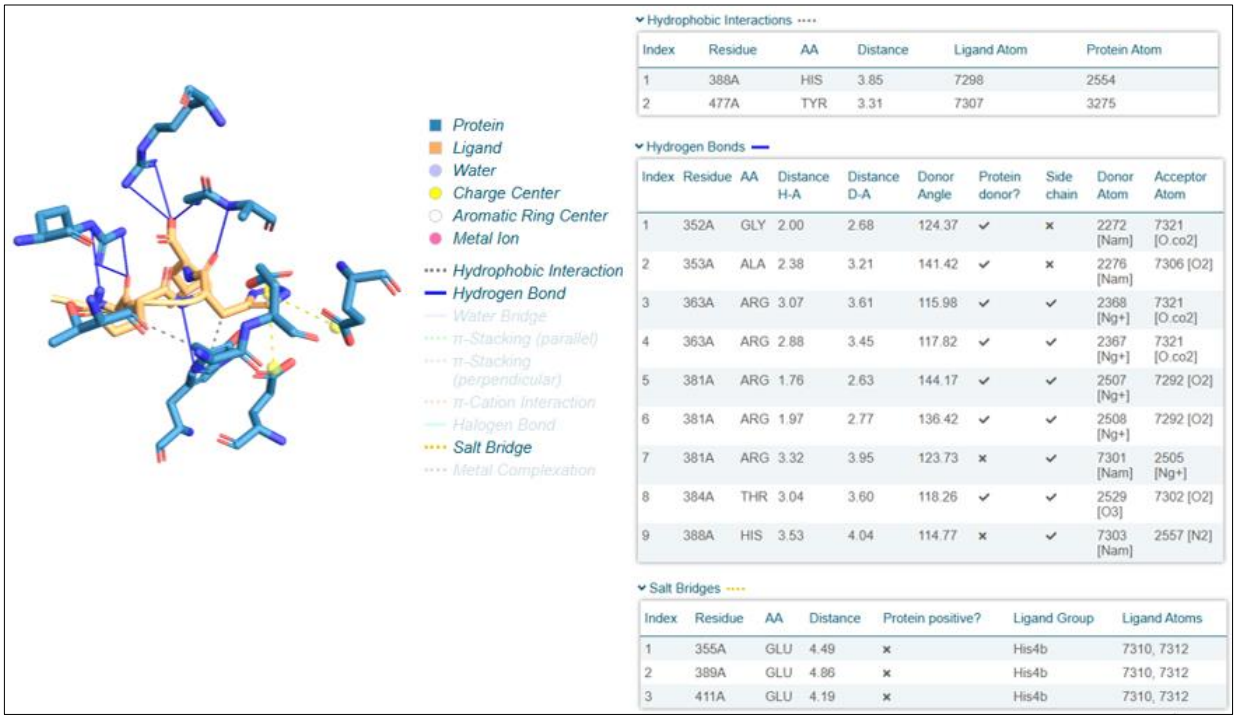
Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	442A	ARG	3.74	7307	3017
2	706A	LYS	3.21	7352	5153
3	722A	LEU	3.98	7351	5290
4	864A	ILE	2.75	7314	6443
5	864A	ILE	3.03	7315	6444
6	900A	ASN	3.47	7315	6727
7	904A	ALA	3.45	7315	6760

Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	381A	ARG	2.12	2.74	120.10	✓	✓	2507 [Ng+]	7293 [O2]
2	418A	GLU	3.26	3.96	127.37	×	✓	7290 [N3]	2820 [O3]
3	437A	LEU	3.09	4.01	156.57	×	×	7325 [Ng+]	2972 [O2]
4	442A	ARG	3.07	3.83	135.41	✓	✓	3021 [Ng+]	7301 [O2]
5	703A	SER	2.30	3.12	142.04	✓	✓	5123 [O3]	7330 [O2]
6	758A	SER	3.26	3.91	126.08	✓	✓	5597 [O3]	7348 [O.co2]
7	762A	SER	3.28	4.09	141.45	✓	✓	5621 [O3]	7353 [O.co2]
8	903A	GLN	2.16	2.73	115.52	✓	✓	6754 [Nam]	7311 [O2]
9	907A	ARG	3.34	3.81	111.38	×	✓	7327 [Nam]	6784 [Ng+]
10	907A	ARG	3.45	4.01	118.59	✓	✓	6785 [Ng+]	7319 [O2]
11	907A	ARG	2.48	3.33	143.64	✓	✓	6784 [Ng+]	7311 [O2]

Secuencia 75: LAQHL



Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	388A	HIS	3.85	7298	2554
2	477A	TYR	3.31	7307	3275

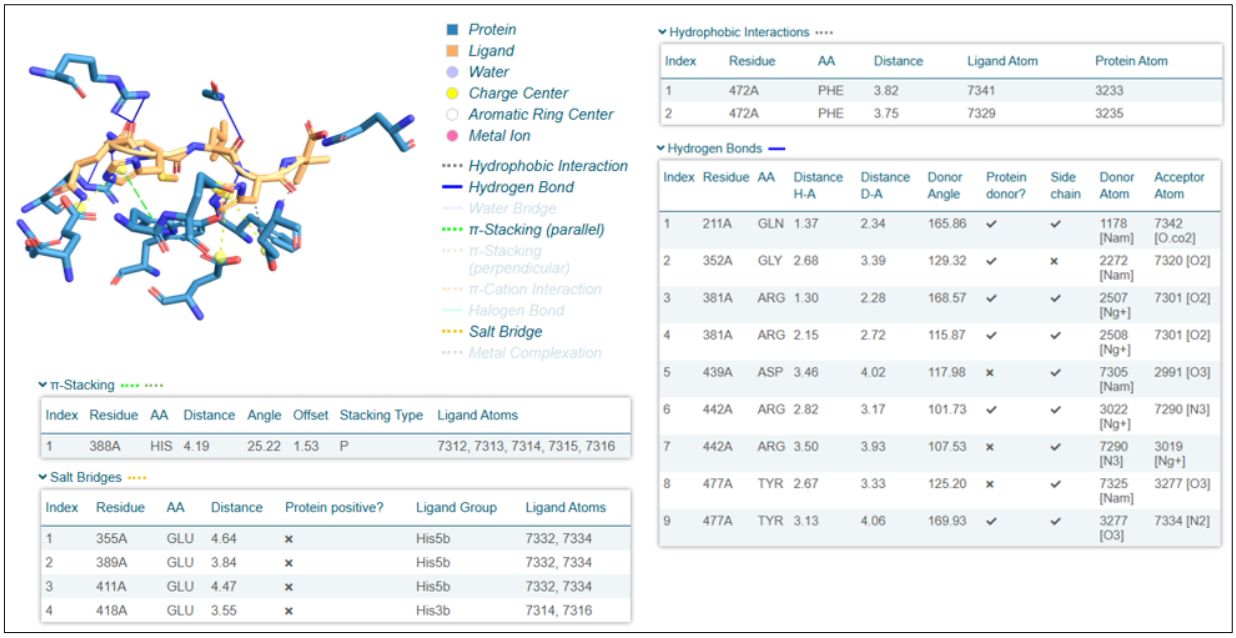
Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	352A	GLY	2.00	2.68	124.37	✓	×	2272 [Nam]	7321 [O.co2]
2	353A	ALA	2.38	3.21	141.42	✓	×	2276 [Nam]	7306 [O2]
3	363A	ARG	3.07	3.61	115.98	✓	✓	2368 [Ng+]	7321 [O.co2]
4	363A	ARG	2.88	3.45	117.82	✓	✓	2367 [Ng+]	7321 [O.co2]
5	381A	ARG	1.76	2.63	144.17	✓	✓	2507 [Ng+]	7292 [O2]
6	381A	ARG	1.97	2.77	136.42	✓	✓	2508 [Ng+]	7292 [O2]
7	381A	ARG	3.32	3.95	123.73	×	✓	7301 [Nam]	2505 [Ng+]
8	384A	THR	3.04	3.60	118.26	✓	✓	2529 [O3]	7302 [O2]
9	388A	HIS	3.53	4.04	114.77	×	✓	7303 [Nam]	2557 [N2]

Salt Bridges

Index	Residue	AA	Distance	Protein positive?	Ligand Group	Ligand Atoms
1	355A	GLU	4.49	×	His4b	7310, 7312
2	389A	GLU	4.86	×	His4b	7310, 7312
3	411A	GLU	4.19	×	His4b	7310, 7312

Secuencia 78: MQHLHV



Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	472A	PHE	3.82	7341	3233
2	472A	PHE	3.75	7329	3235

Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	211A	GLN	1.37	2.34	165.86	✓	✓	1178 [Nam]	7342 [O.co2]
2	352A	GLY	2.68	3.39	129.32	✓	×	2272 [Nam]	7320 [O2]
3	381A	ARG	1.30	2.28	168.57	✓	✓	2507 [Ng+]	7301 [O2]
4	381A	ARG	2.15	2.72	115.87	✓	✓	2508 [Ng+]	7301 [O2]
5	439A	ASP	3.46	4.02	117.98	×	✓	7305 [Nam]	2991 [O3]
6	442A	ARG	2.82	3.17	101.73	✓	✓	3022 [Ng+]	7290 [N3]
7	442A	ARG	3.50	3.93	107.53	×	✓	7290 [N3]	3019 [Ng+]
8	477A	TYR	2.67	3.33	125.20	×	✓	7325 [Nam]	3277 [O3]
9	477A	TYR	3.13	4.06	169.93	✓	✓	3277 [O3]	7334 [N2]

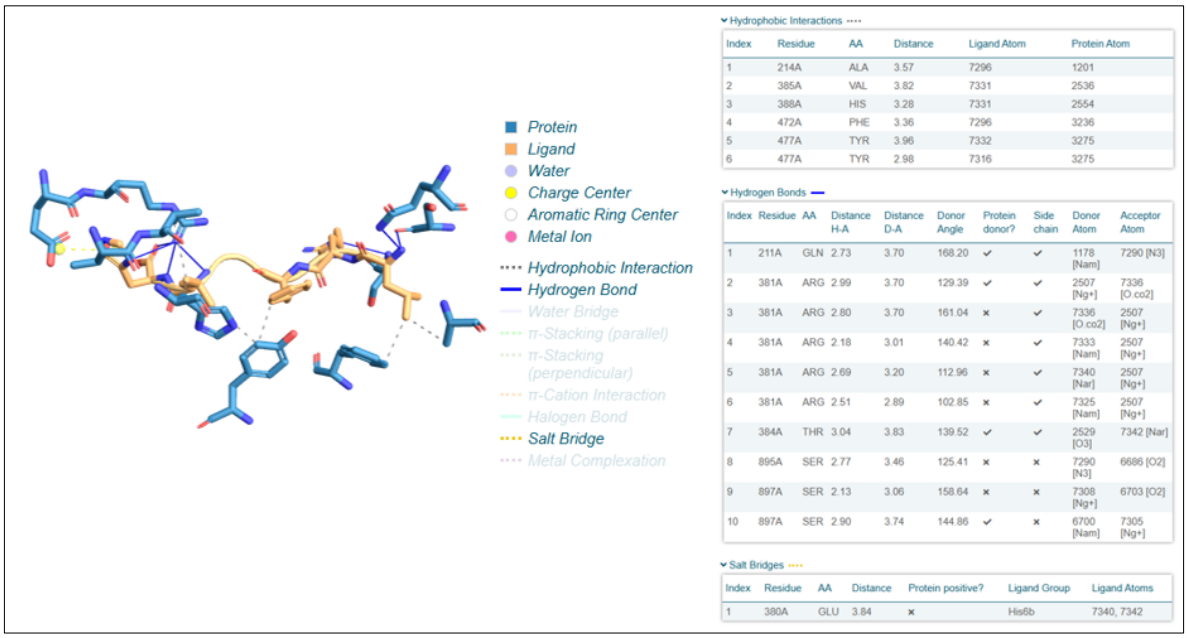
π-Stacking

Index	Residue	AA	Distance	Angle	Offset	Stacking Type	Ligand Atoms
1	388A	HIS	4.19	25.22	1.53	P	7312, 7313, 7314, 7315, 7316

Salt Bridges

Index	Residue	AA	Distance	Protein positive?	Ligand Group	Ligand Atoms
1	355A	GLU	4.64	×	His5b	7332, 7334
2	389A	GLU	3.84	×	His5b	7332, 7334
3	411A	GLU	4.47	×	His5b	7332, 7334
4	418A	GLU	3.55	×	His3b	7314, 7316

Secuencia 89: LRFALH



Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	214A	ALA	3.57	7296	1201
2	385A	VAL	3.82	7331	2536
3	388A	HIS	3.28	7331	2554
4	472A	PHE	3.36	7296	3236
5	477A	TYR	3.96	7332	3275
6	477A	TYR	2.98	7316	3275

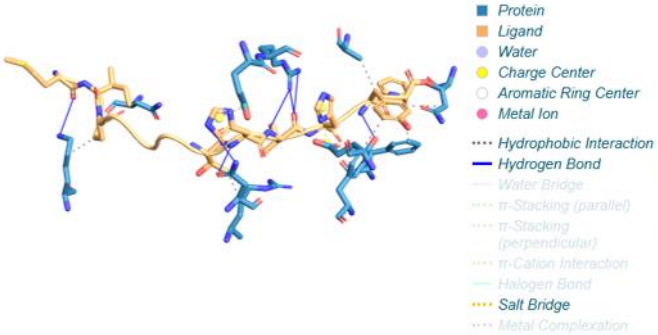
Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	211A	GLN	2.73	3.70	168.20	✓	✓	1178 [Nam]	7290 [N3]
2	381A	ARG	2.99	3.70	129.39	✓	✓	2507 [Ng+]	7336 [O.co2]
3	381A	ARG	2.80	3.70	161.04	×	✓	7336 [O.co2]	2507 [Ng+]
4	381A	ARG	2.18	3.01	140.42	×	✓	7333 [Nam]	2507 [Ng+]
5	381A	ARG	2.69	3.20	112.96	×	✓	7340 [Nar]	2507 [Ng+]
6	381A	ARG	2.51	2.89	102.85	×	✓	7325 [Nam]	2507 [Ng+]
7	384A	THR	3.04	3.83	139.52	✓	✓	2529 [O3]	7342 [Nar]
8	895A	SER	2.77	3.46	125.41	×	×	7290 [N3]	6686 [O2]
9	897A	SER	2.13	3.06	158.64	×	×	7308 [Ng+]	6703 [O2]
10	897A	SER	2.90	3.74	144.86	✓	×	6700 [Nam]	7305 [Ng+]

Salt Bridges

Index	Residue	AA	Distance	Protein positive?	Ligand Group	Ligand Atoms
1	380A	GLU	3.84	×	His6b	7340, 7342

Secuencia 90: MIPFLATHIHIYF



Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	213A	GLN	3.90	7396	1193
2	351A	ALA	3.70	7384	2271
3	442A	ARG	3.49	7342	3017
4	472A	PHE	3.71	7396	3237
5	477A	TYR	3.79	7388	3275
6	706A	LYS	3.43	7311	5153
7	868A	ASN	3.59	7305	6470

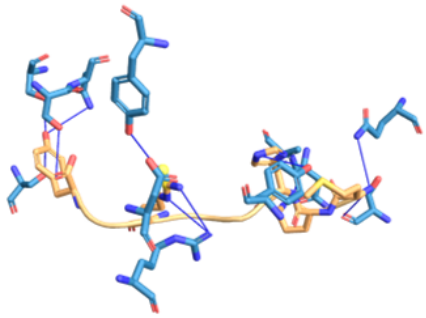
Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	381A	ARG	3.47	4.10	123.62	×	✓	7354 [Nam]	2507 [Ng+]
2	381A	ARG	1.43	2.31	146.12	✓	✓	2507 [Ng+]	7357 [O2]
3	381A	ARG	3.17	3.65	111.94	✓	✓	2508 [Ng+]	7357 [O2]
4	438A	ASN	2.34	3.27	157.02	×	✓	7344 [Nam]	2984 [O2]
5	438A	ASN	2.97	3.86	151.05	✓	✓	2983 [Nam]	7353 [N2]
6	477A	TYR	1.79	2.74	162.53	×	✓	7380 [Nam]	3277 [O3]
7	706A	LYS	3.30	3.92	120.20	✓	✓	5155 [N3+]	7293 [O2]

Salt Bridges

Index	Residue	AA	Distance	Protein positive?	Ligand Group	Ligand Atoms
1	380A	GLU	3.62	×	His8b	7351, 7353
2	418A	GLU	3.65	×	His10b	7369, 7371

Secuencia 97: MQHLLRYY



Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	351A	ALA	3.47	7311	2271

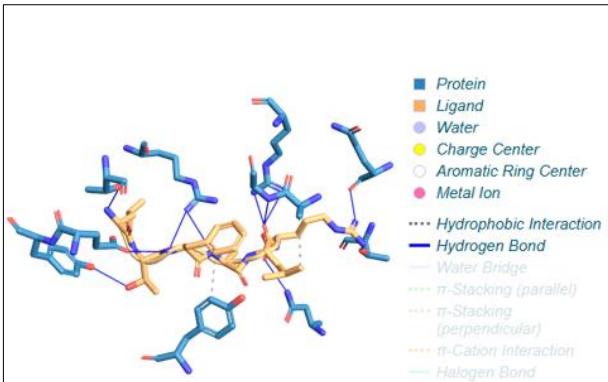
Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	211A	GLN	3.16	4.06	154.03	✓	✓	1178 [Nam]	7290 [N3]
2	350A	ASN	2.14	2.75	118.55	×	✓	7305 [Nam]	2266 [O2]
3	350A	ASN	3.29	3.93	124.25	×	×	7307 [Nam]	2262 [O2]
4	375A	SER	1.97	2.84	148.47	✓	✓	2459 [O3]	7367 [O3]
5	375A	SER	2.32	2.84	115.04	×	✓	7367 [O3]	2459 [O3]
6	377A	SER	3.17	3.86	129.76	✓	✓	2471 [O3]	7359 [O.co2]
7	378A	ASN	3.57	4.02	110.53	✓	✓	2478 [Nam]	7367 [O3]
8	419A	TYR	2.73	3.43	128.34	×	✓	7342 [Ng+]	2833 [O3]
9	442A	ARG	3.12	3.87	134.13	✓	✓	3021 [Ng+]	7340 [Ng+]
10	442A	ARG	2.94	3.86	157.39	×	✓	7343 [Ng+]	3021 [Ng+]
11	477A	TYR	3.49	3.88	107.80	✓	✓	3277 [O3]	7316 [Nar]
12	794A	SER	3.38	3.83	110.80	✓	✓	5877 [O3]	7367 [O3]
13	895A	SER	3.14	4.02	144.29	×	×	7290 [N3]	6686 [O2]

Salt Bridges

Index	Residue	AA	Distance	Protein positive?	Ligand Group	Ligand Atoms
1	439A	ASP	4.04	×	Arg6b	7340, 7342, 7343

Secuencia 98: LTFALR



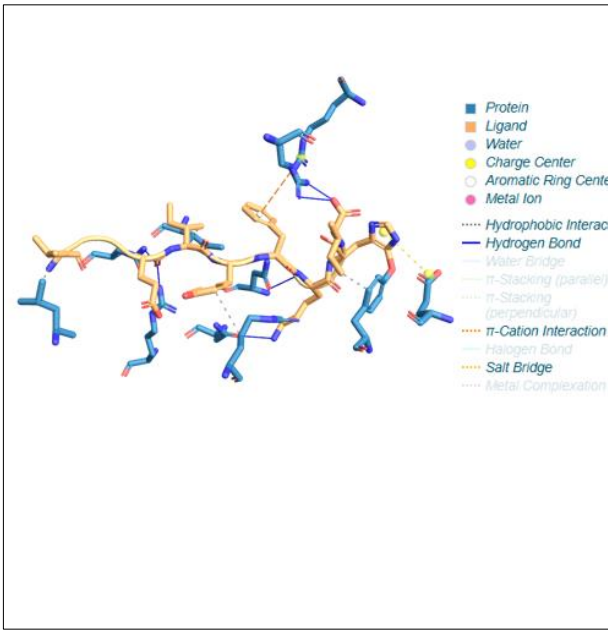
Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	351A	ALA	3.81	7327	2271
2	477A	TYR	3.39	7311	3275

Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	352A	GLY	2.58	3.11	113.93	✓	×	2272 [Nam]	7324 [O2]
2	363A	ARG	2.53	3.14	120.63	✓	✓	2367 [Ng+]	7324 [O2]
3	363A	ARG	2.35	3.00	123.17	✓	✓	2368 [Ng+]	7324 [O2]
4	381A	ARG	2.14	3.12	173.93	✓	✓	2507 [Ng+]	7301 [O2]
5	381A	ARG	3.00	3.77	135.69	×	✓	7316 [Nam]	2507 [Ng+]
6	384A	THR	1.96	2.86	152.33	✓	✓	2529 [O3]	7290 [N3]
7	384A	THR	1.88	2.86	160.59	×	✓	7290 [N3]	2529 [O3]
8	418A	GLU	2.99	3.93	161.10	×	✓	7305 [Nam]	2820 [O3]
9	419A	TYR	2.66	3.22	116.61	×	✓	7304 [O3]	2833 [O3]
10	857A	GLN	1.80	2.55	129.99	×	×	7338 [Ng+]	6390 [O2]
11	860A	THR	3.60	3.97	104.98	×	✓	7339 [Ng+]	6415 [O3]
12	900A	ASN	2.75	3.27	113.54	✓	✓	6729 [Nam]	7332 [O.co2]

Secuencia 99: LMEIYFKHI



Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	442A	ARG	3.99	7332	3017
2	477A	TYR	2.71	7372	3273
3	699A	LEU	3.18	7296	5096
4	864A	ILE	3.84	7321	6443

Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	381A	ARG	2.51	3.28	135.15	✓	✓	2507 [Ng+]	7373 [O.co2]
2	381A	ARG	1.73	2.44	126.51	✓	✓	2508 [Ng+]	7373 [O.co2]
3	868A	ASN	2.64	3.18	114.81	✓	✓	6472 [Nam]	7309 [O2]
4	900A	ASN	3.35	3.93	119.50	×	✓	7346 [Nam]	6730 [O2]
5	903A	GLN	2.14	3.01	141.78	×	✓	7354 [N3+]	6755 [O2]
6	907A	ARG	2.42	3.05	121.29	✓	✓	6785 [Ng+]	7309 [O2]

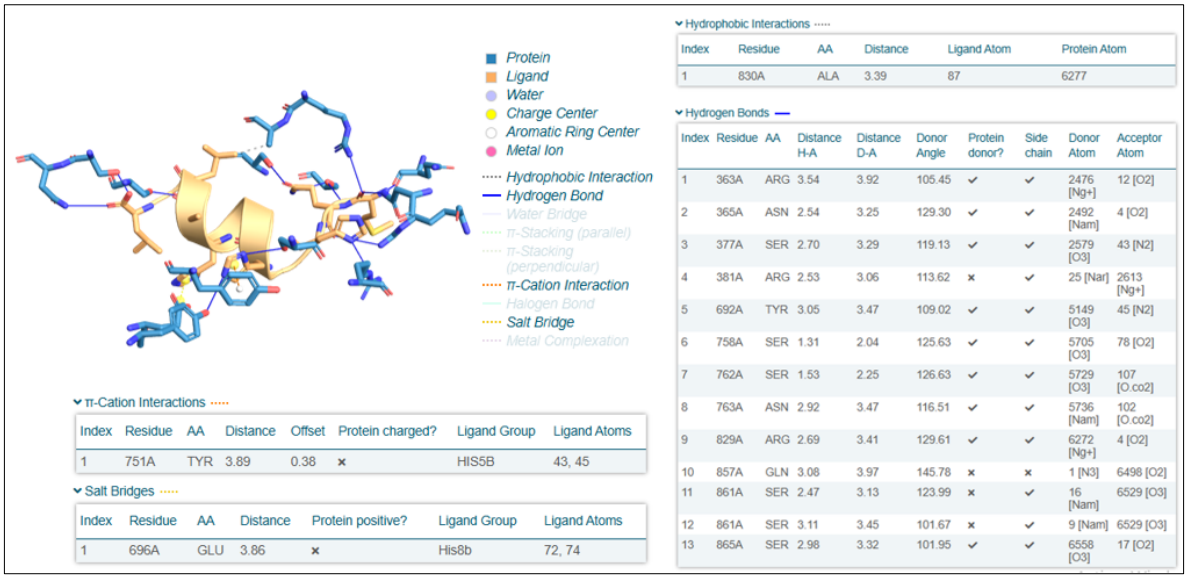
π-Cation Interactions

Index	Residue	AA	Distance	Offset	Protein charged?	Ligand Group	Ligand Atoms
1	363A	ARG	5.95	0.73	✓	Aromatic	7340, 7341, 7342, 7343, 7344, 7345

Salt Bridges

Index	Residue	AA	Distance	Protein positive?	Ligand Group	Ligand Atoms
1	411A	GLU	4.81	×	His6b	7362, 7364

Secuencia 104: MQHLHFLHALFL



Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	830A	ALA	3.39	87	6277

Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	363A	ARG	3.54	3.92	105.45	✓	✓	2476 [Ng+]	12 [O2]
2	365A	ASN	2.54	3.25	129.30	✓	✓	2492 [Nam]	4 [O2]
3	377A	SER	2.70	3.29	119.13	✓	✓	2579 [O3]	43 [N2]
4	381A	ARG	2.53	3.06	113.62	×	✓	25 [Nar]	2613 [Ng+]
5	692A	TYR	3.05	3.47	109.02	✓	✓	5149 [O3]	45 [N2]
6	758A	SER	1.31	2.04	125.63	✓	✓	5705 [O3]	78 [O2]
7	762A	SER	1.53	2.25	126.63	✓	✓	5729 [O3]	107 [O.co2]
8	763A	ASN	2.92	3.47	116.51	✓	✓	5736 [Nam]	102 [O.co2]
9	829A	ARG	2.69	3.41	129.61	✓	✓	6272 [Ng+]	4 [O2]
10	857A	GLN	3.08	3.97	145.78	×	×	1 [N3]	6498 [O2]
11	861A	SER	2.47	3.13	123.99	×	✓	16 [Nam]	6529 [O3]
12	861A	SER	3.11	3.45	101.67	×	✓	9 [Nam]	6529 [O3]
13	865A	SER	2.98	3.32	101.95	✓	✓	6558 [O3]	17 [O2]

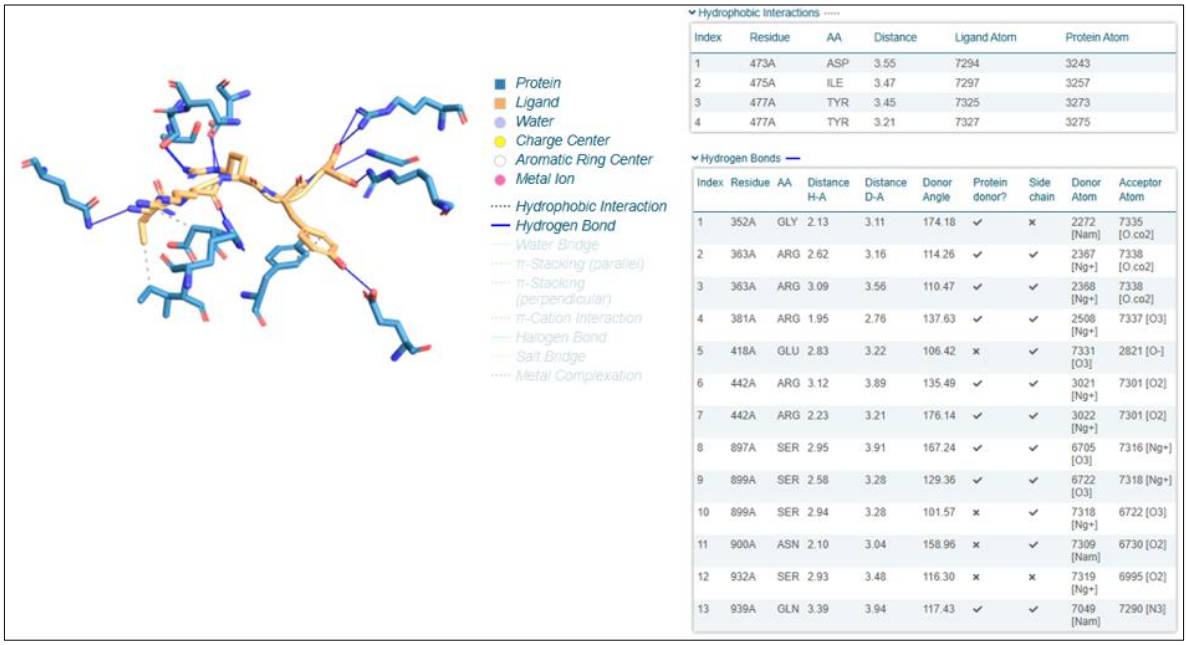
π-Cation Interactions

Index	Residue	AA	Distance	Offset	Protein charged?	Ligand Group	Ligand Atoms
1	751A	TYR	3.89	0.38	×	HIS5B	43, 45

Salt Bridges

Index	Residue	AA	Distance	Protein positive?	Ligand Group	Ligand Atoms
1	696A	GLU	3.86	×	His8b	72, 74

Secuencia 113: LRRYS



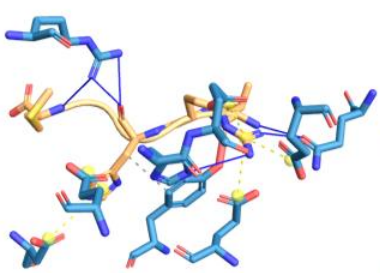
Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	473A	ASP	3.55	7294	3243
2	475A	ILE	3.47	7297	3257
3	477A	TYR	3.45	7325	3273
4	477A	TYR	3.21	7327	3275

Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	352A	GLY	2.13	3.11	174.18	✓	×	2272 [Nam]	7335 [O.co2]
2	363A	ARG	2.62	3.16	114.26	✓	✓	2367 [Ng+]	7338 [O.co2]
3	363A	ARG	3.09	3.56	110.47	✓	✓	2368 [Ng+]	7338 [O.co2]
4	381A	ARG	1.95	2.76	137.63	✓	✓	2508 [Ng+]	7337 [O3]
5	418A	GLU	2.83	3.22	106.42	×	✓	7331 [O3]	2821 [O-]
6	442A	ARG	3.12	3.89	135.49	✓	✓	3021 [Ng+]	7301 [O2]
7	442A	ARG	2.23	3.21	176.14	✓	✓	3022 [Ng+]	7301 [O2]
8	897A	SER	2.95	3.91	167.24	✓	✓	6705 [O3]	7316 [Ng+]
9	899A	SER	2.58	3.28	129.36	✓	✓	6722 [O3]	7318 [Ng+]
10	899A	SER	2.94	3.28	101.57	×	✓	7318 [Ng+]	6722 [O3]
11	900A	ASN	2.10	3.04	158.96	×	✓	7309 [Nam]	6730 [O2]
12	932A	SER	2.93	3.48	116.30	×	×	7319 [Ng+]	6995 [O2]
13	939A	GLN	3.39	3.94	117.43	✓	✓	7049 [Nam]	7290 [N3]

Secuencia 124: LRHLM



Protein
Ligand
Water
Charge Center
Aromatic Ring Center
Metal Ion
Hydrophobic Interaction
Hydrogen Bond
Water Bridge
π-Stacking (parallel)
π-Stacking (perpendicular)
π-Cation Interaction
Halogen Bond
Salt Bridge
Metal Complexation

Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	477A	TYR	3.87	7313	3273

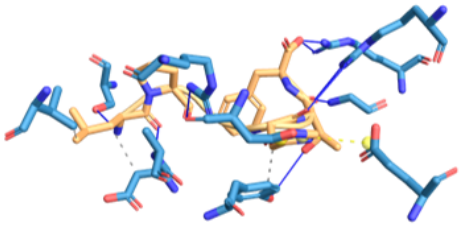
Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	213A	GLN	2.76	3.56	135.21	×	✓	7290 [N3]	1196 [O2]
2	213A	GLN	3.21	3.77	117.72	×	✓	7307 [Ng+]	1196 [O2]
3	381A	ARG	2.83	3.72	150.77	×	✓	7327 [Nam]	2507 [Ng+]
4	381A	ARG	1.60	2.55	159.69	✓	✓	2507 [Ng+]	7312 [O2]
5	381A	ARG	2.52	3.20	126.57	✓	✓	2508 [Ng+]	7312 [O2]
6	388A	HIS	3.10	3.73	122.37	×	✓	7308 [Ng+]	2559 [N2]

Salt Bridges

Index	Residue	AA	Distance	Protein positive?	Ligand Group	Ligand Atoms
1	355A	GLU	3.16	×	Arg2b	7305, 7307, 7308
2	389A	GLU	3.56	×	Arg2b	7305, 7307, 7308
3	411A	GLU	3.91	×	Arg2b	7305, 7307, 7308
4	418A	GLU	5.33	×	His3b	7316, 7318
5	439A	ASP	4.59	×	His3b	7316, 7318

Secuencia 135: LPGHTF



Protein
Ligand
Water
Charge Center
Aromatic Ring Center
Metal Ion
Hydrophobic Interaction
Hydrogen Bond
Water Bridge
π-Stacking (parallel)
π-Stacking (perpendicular)
π-Cation Interaction
Halogen Bond
Salt Bridge
Metal Complexation

Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	446A	VAL	3.23	7287	3049
2	473A	ASP	3.52	7285	3243
3	477A	TYR	2.76	7304	3273

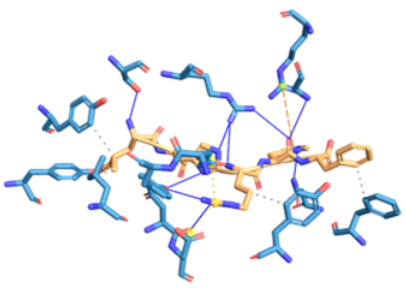
Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	352A	GLY	2.37	3.02	123.41	✓	×	2272 [Nam]	7313 [O2]
2	363A	ARG	1.32	2.11	131.79	✓	✓	2368 [Ng+]	7328 [O.co2]
3	363A	ARG	3.36	3.75	105.49	✓	✓	2367 [Ng+]	7328 [O.co2]
4	381A	ARG	3.05	3.95	152.15	✓	✓	2507 [Ng+]	7303 [O2]
5	381A	ARG	2.22	3.05	140.63	✓	✓	2508 [Ng+]	7313 [O2]
6	442A	ARG	2.37	3.18	138.67	✓	✓	3021 [Ng+]	7292 [O2]
7	442A	ARG	1.97	2.89	154.61	✓	✓	3022 [Ng+]	7292 [O2]
8	474A	ALA	3.04	3.52	111.30	✓	×	3247 [Nam]	7284 [O2]
9	477A	TYR	3.22	3.77	120.04	✓	✓	3277 [O3]	7316 [O3]
10	899A	SER	2.67	3.45	138.00	✓	✓	6713 [O3]	7281 [N3]
11	899A	SER	2.88	3.45	116.23	×	✓	7281 [N3]	6713 [O3]

Salt Bridges

Index	Residue	AA	Distance	Protein positive?	Ligand Group	Ligand Atoms
1	418A	GLU	3.83	×	His4b	7307, 7309
2	439A	ASP	3.92	×	His4b	7307, 7309

Secuencia 141: LHLRHF



Protein
Ligand
Water
Charge Center
Aromatic Ring Center
Metal Ion
Hydrophobic Interaction
Hydrogen Bond
Water Bridge
π-Stacking (parallel)
π-Stacking (perpendicular)
π-Cation Interaction
Halogen Bond
Salt Bridge
Metal Complexation

Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	434A	LEU	2.36	7288	2953
2	472A	PHE	3.44	7335	3235
3	477A	TYR	3.30	7312	3273
4	692A	TYR	3.34	7287	5030
5	751A	TYR	3.51	7286	5537

Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	352A	GLY	2.94	3.79	145.38	✓	×	2272 [Nam]	7321 [O2]
2	377A	SER	3.02	3.88	142.43	×	✓	7281 [N3]	2471 [O3]
3	381A	ARG	2.84	3.69	143.89	×	✓	7307 [Nam]	2507 [Ng+]
4	381A	ARG	3.58	3.97	106.41	✓	✓	2507 [Ng+]	7298 [Nar]
5	381A	ARG	3.22	4.10	149.48	✓	✓	2508 [Ng+]	7321 [O2]
6	419A	TYR	3.25	3.61	103.04	×	✓	7296 [Nar]	2833 [O3]
7	419A	TYR	2.32	3.29	166.44	×	✓	7317 [Ng+]	2833 [O3]
8	438A	ASN	2.72	3.56	143.55	×	×	7316 [Ng+]	2980 [O2]
9	900A	ASN	3.00	3.34	101.98	✓	✓	6720 [Nam]	7327 [Nar]

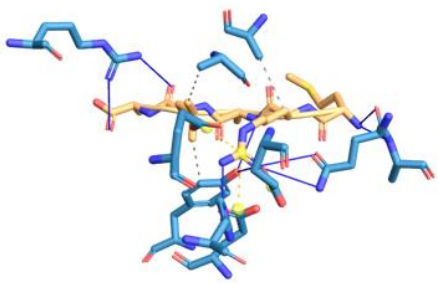
π-Cation Interactions

Index	Residue	AA	Distance	Offset	Protein charged?	Ligand Group	Ligand Atoms
1	363A	ARG	5.84	1.17	✓	Aromatic	7323, 7324, 7325, 7326, 7327

Salt Bridges

Index	Residue	AA	Distance	Protein positive?	Ligand Group	Ligand Atoms
1	418A	GLU	4.05	×	Arg4b	7314, 7316, 7317
2	418A	GLU	3.02	×	His2b	7296, 7298
3	439A	ASP	3.13	×	Arg4b	7314, 7316, 7317

Secuencia 142: MRALA



Protein
Ligand
Water
Charge Center
Aromatic Ring Center
Metal Ion
Hydrophobic Interaction
Hydrogen Bond
Water Bridge
π-Stacking (parallel)
π-Stacking (perpendicular)
π-Cation Interaction
Halogen Bond
Salt Bridge
Metal Complexation

Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	351A	ALA	3.73	7293	2271
2	353A	ALA	3.98	7311	2280
3	477A	TYR	3.60	7312	3275

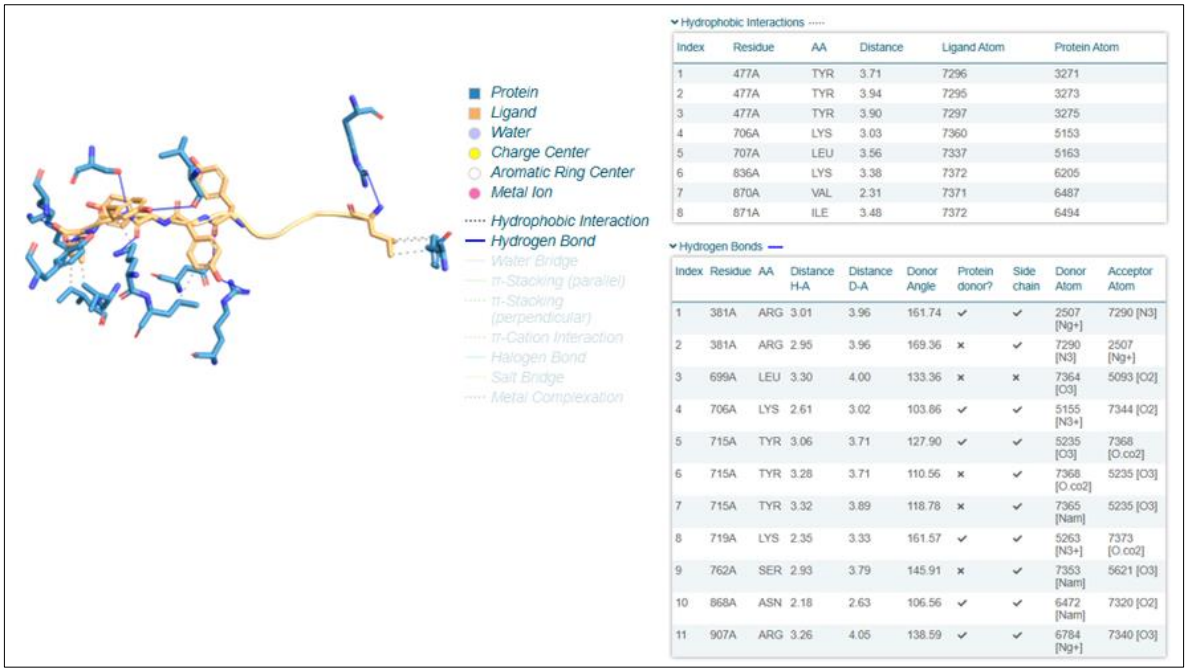
Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	213A	GLN	3.08	3.62	113.74	×	×	7281 [N3]	1191 [O2]
2	213A	GLN	3.16	3.96	138.60	✓	✓	1195 [Nam]	7298 [Ng+]
3	213A	GLN	2.78	3.73	161.12	×	✓	7298 [Ng+]	1196 [O2]
4	214A	ALA	2.82	3.54	129.87	✓	×	1197 [Nam]	7281 [N3]
5	381A	ARG	3.13	3.84	129.99	✓	✓	2507 [Ng+]	7316 [O.co2]
6	381A	ARG	3.12	3.97	146.03	✓	✓	2508 [Ng+]	7308 [O2]
7	392A	HIS	1.78	2.75	166.07	×	✓	7299 [Ng+]	2591 [Nar]
8	477A	TYR	3.23	3.89	125.97	×	✓	7300 [Nam]	3277 [O3]
9	477A	TYR	3.20	3.87	130.35	✓	✓	3277 [O3]	7298 [Ng+]

Salt Bridges

Index	Residue	AA	Distance	Protein positive?	Ligand Group	Ligand Atoms
1	355A	GLU	3.49	×	Arg2b	7296, 7298, 7299
2	389A	GLU	3.93	×	Arg2b	7296, 7298, 7299
3	411A	GLU	3.83	×	Arg2b	7296, 7298, 7299

Secuencia 147: LAKAYYYYI



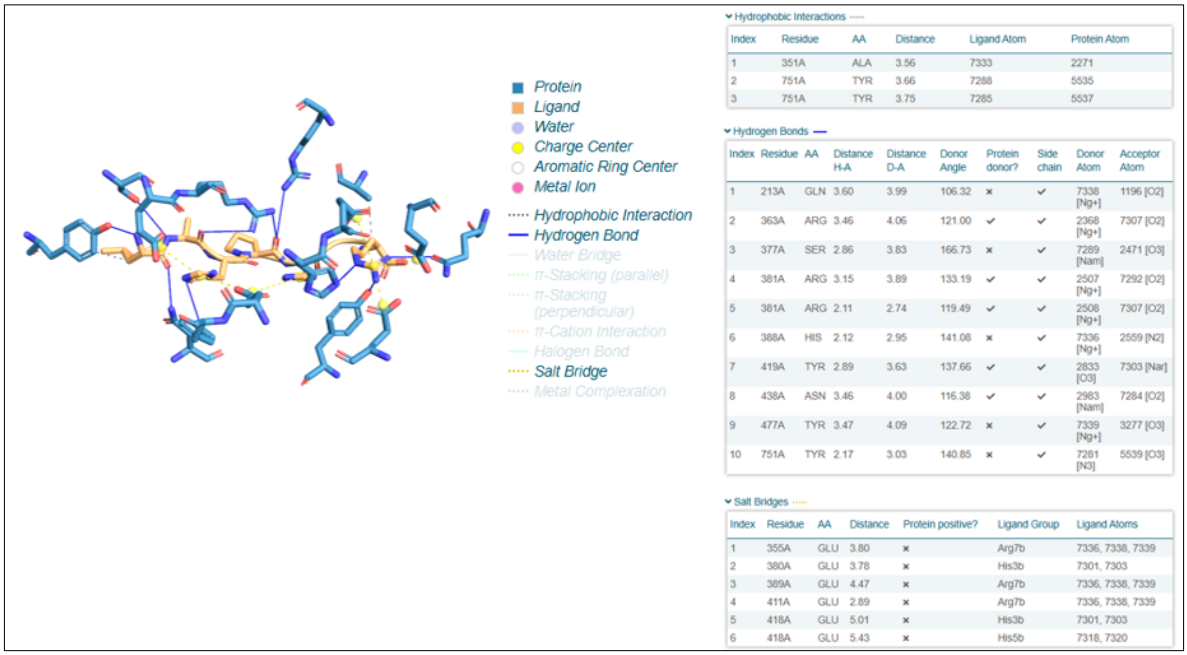
Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	477A	TYR	3.71	7296	3271
2	477A	TYR	3.94	7295	3273
3	477A	TYR	3.90	7297	3275
4	706A	LYS	3.03	7360	5153
5	707A	LEU	3.56	7337	5163
6	836A	LYS	3.38	7372	6205
7	870A	VAL	2.31	7371	6487
8	871A	ILE	3.48	7372	6494

Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	381A	ARG	3.01	3.96	161.74	✓	✓	2507 [Ng+]	7290 [N3]
2	381A	ARG	2.95	3.96	169.36	×	✓	7290 [N3]	2507 [Ng+]
3	699A	LEU	3.30	4.00	133.36	×	×	7364 [O3]	5093 [O2]
4	706A	LYS	2.61	3.02	103.86	✓	✓	5155 [N3+]	7344 [O2]
5	715A	TYR	3.06	3.71	127.90	✓	✓	5235 [O3]	7368 [O.co2]
6	715A	TYR	3.28	3.71	110.56	×	✓	7368 [O.co2]	5235 [O3]
7	715A	TYR	3.32	3.89	118.78	×	✓	7365 [Nam]	5235 [O3]
8	719A	LYS	2.35	3.33	161.57	✓	✓	5263 [N3+]	7373 [O.co2]
9	762A	SER	2.93	3.79	145.91	×	✓	7353 [Nam]	5621 [O3]
10	868A	ASN	2.18	2.63	106.56	✓	✓	6472 [Nam]	7320 [O2]
11	907A	ARG	3.26	4.05	138.59	✓	✓	6784 [Ng+]	7340 [O3]

Secuencia 149: LAHPHLR



Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	351A	ALA	3.56	7333	2271
2	751A	TYR	3.66	7288	5535
3	751A	TYR	3.75	7285	5537

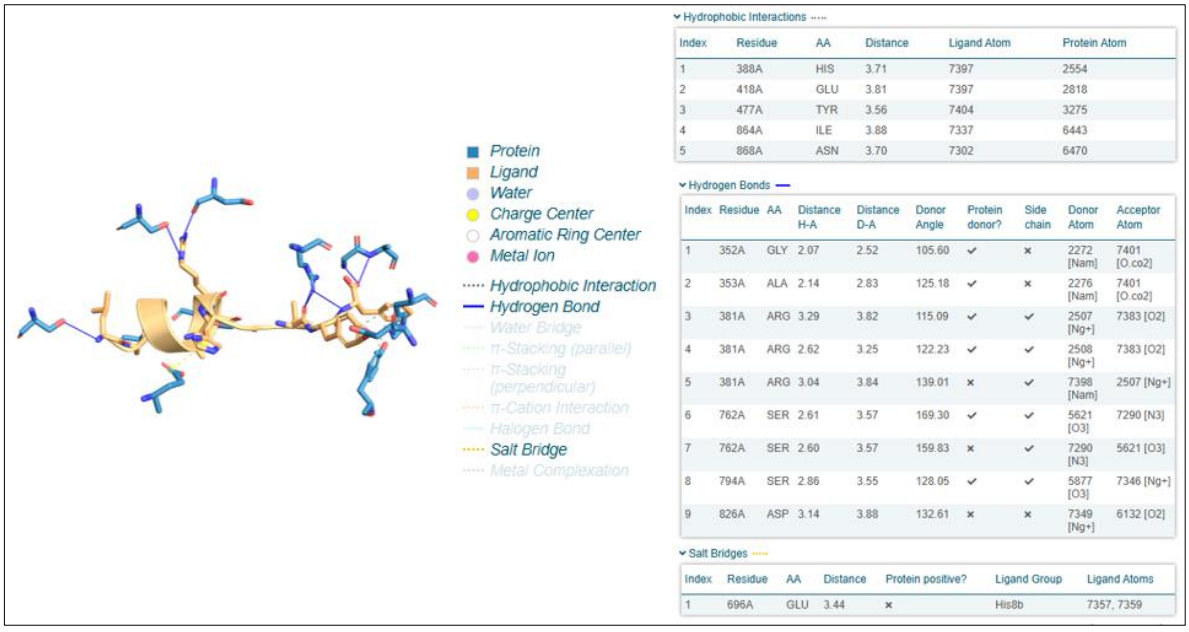
Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	213A	GLN	3.60	3.99	106.32	×	✓	7338 [Ng+]	1196 [O2]
2	363A	ARG	3.46	4.06	121.00	✓	✓	2368 [Ng+]	7307 [O2]
3	377A	SER	2.86	3.83	166.73	×	✓	7289 [Nam]	2471 [O3]
4	381A	ARG	3.15	3.89	133.19	✓	✓	2507 [Ng+]	7292 [O2]
5	381A	ARG	2.11	2.74	119.49	✓	✓	2508 [Ng+]	7307 [O2]
6	388A	HIS	2.12	2.95	141.08	×	✓	7336 [Ng+]	2559 [N2]
7	419A	TYR	2.89	3.63	137.66	✓	✓	2833 [O3]	7303 [Nar]
8	438A	ASN	3.46	4.00	116.38	✓	✓	2983 [Nam]	7284 [O2]
9	477A	TYR	3.47	4.09	122.72	×	✓	7339 [Ng+]	3277 [O3]
10	751A	TYR	2.17	3.03	140.85	×	✓	7261 [N3]	5539 [O3]

Salt Bridges

Index	Residue	AA	Distance	Protein positive?	Ligand Group	Ligand Atoms
1	355A	GLU	3.80	×	Arg7b	7336, 7338, 7339
2	380A	GLU	3.78	×	His3b	7301, 7303
3	389A	GLU	4.47	×	Arg7b	7336, 7338, 7339
4	411A	GLU	2.89	×	Arg7b	7336, 7338, 7339
5	418A	GLU	5.01	×	His3b	7301, 7303
6	418A	GLU	5.43	×	His5b	7318, 7320

Secuencia 157: LAHHLLRHHHVFF



Hydrophobic Interactions

Index	Residue	AA	Distance	Ligand Atom	Protein Atom
1	388A	HIS	3.71	7397	2554
2	418A	GLU	3.81	7397	2818
3	477A	TYR	3.56	7404	3275
4	864A	ILE	3.88	7337	6443
5	868A	ASN	3.70	7302	6470

Hydrogen Bonds

Index	Residue	AA	Distance H-A	Distance D-A	Donor Angle	Protein donor?	Side chain	Donor Atom	Acceptor Atom
1	352A	GLY	2.07	2.52	105.60	✔	×	2272 [Nam]	7401 [O.co2]
2	353A	ALA	2.14	2.83	125.18	✔	×	2276 [Nam]	7401 [O.co2]
3	381A	ARG	3.29	3.82	115.09	✔	✔	2507 [Ng+]	7383 [O2]
4	381A	ARG	2.62	3.25	122.23	✔	✔	2508 [Ng+]	7383 [O2]
5	381A	ARG	3.04	3.84	139.01	×	✔	7398 [Nam]	2507 [Ng+]
6	762A	SER	2.61	3.57	169.30	✔	✔	5621 [O3]	7290 [N3]
7	762A	SER	2.60	3.57	159.83	×	✔	7290 [N3]	5621 [O3]
8	794A	SER	2.86	3.55	128.05	✔	✔	5877 [O3]	7346 [Ng+]
9	826A	ASP	3.14	3.88	132.61	×	×	7349 [Ng+]	6132 [O2]

Salt Bridges

Index	Residue	AA	Distance	Protein positive?	Ligand Group	Ligand Atoms
1	696A	GLU	3.44	×	His8b	7357, 7359